

# Neha R

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## Summary

AI & ML undergraduate specializing in Intelligent Process Automation and NLP for the BFSI (Banking, Financial Services, and Insurance) sector. Experience in architecting Multi-Agent Systems for loan lifecycles and offline OCR pipelines for sensitive document processing. Passionate about deploying responsible, compliant AI solutions to **minimize manual intervention and drive operational excellence** in transaction banking.

## Skills

- **AI/ML & Vision:** PyTorch, Transformers, Hugging Face, TensorFlow, TrOCR, OpenCV, LLM Orchestration
- **Backend & Tools:** Python, FastAPI, Docker, Git, Linux/Unix, SQL
- **Concepts:** Natural Language Processing (NLP), Computer Vision, Multi-Agent Systems, Data Analysis
- **Enterprise Tech:** Apex, LWC, Salesforce Flows

## Projects

**AutoForm-AI – Offline Transformer OCR System** (Python, TrOCR, PyTorch)  Nov 2025

- Engineered a **100% offline AI backend** using **FastAPI**, orchestrating **Microsoft TrOCR** (Transformers) and **EasyOCR** to process sensitive PII data locally without cloud latency.
- Conducted feasibility studies on **Hugging Face** models to optimize inference time for low-resource environments.
- Implemented a **Tamper Detection System** using **Fuzzy Logic** (Levenshtein Distance) to verify on-screen form data against document ground truth, ensuring data validity.
- Designed an advanced computer vision pipeline with **OpenCV** (Morphological Dilation) to segment and extract handwritten text from noisy, low-quality images.

**Agentic Loan Workflow Automation** (Python, FastAPI, Gemini, Docker)  Dec 2025

- Architected a **Multi-Agent AI system** automating the loan lifecycle (Sales → Sanction), reducing processing time by **90%** via autonomous **LLM orchestration**.
- Developed a hybrid **Intent Router** using **FastAPI** to switch between probabilistic LLM dialogue and deterministic underwriting logic, ensuring **100% protocol compliance**.
- Deployed a containerized architecture using **Docker** to decouple the inference engine from the frontend, ensuring scalable deployment of the simulated banking environment.

## Experience

**Salesforce Intern – Impaqtive - An Infogain Company** Oct 2025 – Dec 2025

- Implemented **8+ Automation Flows** (Screen & Record-Triggered), reducing manual steps in lead handling and user onboarding by **20–30%**.
- Configured **data models** (objects, fields, schemas) across 5 modules, optimizing system navigation and data integrity for enterprise users.
- Documented technical configurations and admin workflows across **50+ pages**, facilitating knowledge transfer and onboarding.

## Education

**Vellore Institute of Technology (VIT), Bhopal** June 2022 – April 2027

- Integrated M.Tech in Artificial Intelligence & Machine Learning
- **CGPA: 8.60 / 10.0**
- *Relevant Coursework:* **Linear Algebra, Probability & Statistics**, Deep Learning (Transformers), Data Structures

## Certificates

- Salesforce Developer with Agentblazer Champion (2025)
- Cloud Computing – NPTEL (2024)
- Applied Machine Learning in Python (2023)